

Builder Basics · Pyramid Variations — English Worksheet

Topic: Pyramid Variations · **Course:** Builder Basics · **Time:** about 45 minutes

1 · Predict 🧠

Read each set of steps. Write what you think will happen.

```

set size to 7
repeat size times:
    repeat size times:
        if row is 0 OR row is size-1 OR col is 0 OR col is size-1:
            place block down
        move forward
    [turn and walk back to start of next row]
    
```

Which blocks get placed — all of them, or just some? Where?


```

set size to 9
while size > 0:
    [build one size × size square layer]
    move up by 1
    set size to size - 4
    
```

How is this pyramid different from a normal one? How many layers?


```
set size to 7
while size > 0:
    [build one HOLLOW size × size layer]
    move up by 1
    set size to size - 2
```

Is this pyramid filled or hollow? Could a person walk inside it?

2 · Spot the Bug 🐛

Each code block below is broken. Rewrite it so it works, then explain the fix.

Bug A — The agent is supposed to build a **hollow layer** — only place a block when it is on the edge of the square (not in the middle).

```
repeat size times:
  repeat size times:
    place block down
    move forward
  [turn and walk back to start of next row]
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The agent is supposed to build a **stepped pyramid** where each layer is 4 blocks shorter than the one below.

```
set size to 9
while size > 0:
  [build one size × size square layer]
  move up by 1
  set size to size - 1
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The agent is supposed to build a pyramid with a **gap of 2 blocks** between each layer (the layers float).

```
set size to 7
while size > 0:
    [build one size × size square layer]
    move up by 1
    set size to size - 2
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Show Your Work 📷 🎥

Switch to your homework world. Build **one pyramid variation** — hollow, stepped, two-block, or your own — then give it a **name** and a **one-sentence description**.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.